

Summary

Product engineer and open-source contributor with deep experience across low-latency UIs, infra tooling, and large-scale JS/TS systems. I ship major features into live, business-critical systems by designing for expressiveness, invariants, and additive change.

Operating Principles: end-to-end product development and architecture; 12-factor back-ends; strongly-typed data; low-latency UIs; vertical integration; tooling; defactoring; inlining and colocation; working in public; writing cultures; rejecting the problem.

Languages: TypeScript/JavaScript, Go, C#, Python, Rust, SQL, Shell/Bash, HTML, CSS.

Experience

Maven Securities

Developer

Jan 2024 – Present

TypeScript, React, Node.js, Go, C#/.NET, Postgres, SQL, Prometheus/Grafana

- Designed adaptive fuzzy matcher (two-pointer w/ contiguous-match bias + greedy fallback); matched 500k+ fields/sec (single-threaded, p99 <3μs/field); supported multi-part queries, ranked matches via custom heuristic scoring and CSS-highlighted results.
- Built high-throughput canvas grid for low-latency trading UI (zoom, hit-testing, SVG rasterization, high-DPI); designed to recover seamlessly from GPU driver crashes.
- Implemented OIDC auth proxy in interpreted Go (Yaegi); supported silent refresh, cross-origin embedding, session resilience and complex flows; deployed as a Traefik plugin into Maven's live production trading UI.
- Built sphinx-query, a production-safe observability and state-forensics CLI spanning Postgres, SSH tunnels, Grafana/ClickHouse logs and Prometheus metrics; used to reconstruct user workspaces, inspect deploy state, and triage live trading UI issues.
- Built the multi-version webserver and control plane serving Sphinx, Maven's production trading UI; per-user channel-to-version resolution enabled staged rollouts and instant rollback, with every released version permanently servable.
- Implemented Windows lock-state detection in a C#/.NET service via native Win32 Terminal Services APIs; P/Invoked session queries and decoded binary session structs to reliably detect trader workstation lock state.
- Diagnosed and resolved long-lived production-client reliability issues, including a network-layer memory leak (AbortSignal + ky) and a Chrome speech-synthesis failure that caused traders to lose audio for hours.
- Designed and delivered a recurring 5-day UI engineering training program for interns and graduates.

Freelance Engineer

Freelance & Private Study

Apr 2023 – Jan 2024

Python, Rust, TypeScript

- Adaptively stabilized streaming LLM output using exponential smoothing over queue growth to reduce jitter.
- Built a CLI for managing LLM app configs via declarative YAML; supported interactive prompts, workspace sync, and deployment flows.
- Wrote technical articles on algorithms, linear algebra, and implementing a neural network in Rust.

Software Engineer (Contract) at T. Rowe Price

Oct 2020 – Mar 2023

TypeScript, Python, React, Node.js

- Built synthetic monitoring service using Playwright; evolved from ASCII snapshot testing to DataFrame-style invariant checks against tabular outputs (null detection, bounds, column relationships, sum validation) running against production.
- Rewrote legacy React application into TypeScript; removed Redux-as-cache layer, doubled Lighthouse score (50 → 98), and significantly improved grid testability.
- Improved reliability of Python, Node, and Kotlin services for data ingestion into S3/PostgreSQL; fixed major bugs in shared libraries for logging, tracing, and scheduling.

Software Architect (Contract) at JPMorgan Chase & Co.

Aug 2019 – Oct 2020

TypeScript, React, Argo, Kubernetes

- Re-hired into newly formed Core UI Infrastructure team to build tooling for high-scale development. Developed CI caching infra and Argo workflows that cut E2E test time from hours to minutes.
- Led upgrade of 50+ component library from Material UI v3 → v4 using codemods, visual diffing, and type-checking.
- Contributed to monorepo strategy and shipped DX tools forming the org's "golden path" for frontend dev.

Technical Lead (Contract) at Shell

Dec 2018 – Jun 2019

Python, Jupyter, Spark, Kubernetes

- Brought in by a former colleague to take over a politically sensitive project; aligned global stakeholders, hired a new team, and replaced consultancy build with in-house platform.

- Delivered Spark/Kubernetes-based system (Azure, Docker, Helm, Jupyter) for data workflows; improved pipeline throughput by 4x.
- Built a collaborative culture focused on capability-building; ran weekly technical sessions to engage and upskill Shell engineers.

Application Engineer (Contract) at JPMorgan Chase & Co.

Jun 2017 – Dec 2018

TypeScript, React

- Led core UI work on the rewrite of 'Execute', JPMorgan's single-dealer platform, migrating from Flash to a modular React/TypeScript system under a hard end-of-life deadline.
- Established project governance: created RFC process, authored proposals (entitlements, module loading, CI), and aligned multiple teams across a shared monorepo (200+ packages).
- Built shell and widget tiling system; engineered internal toolchain for ESM/CommonJS/types using Rollup, Babel, and TypeScript. Created Danger plugin to deploy apps/storybooks for interactive PR review.
- Provided technical review and guidance across teams to maintain consistency and architectural integrity at scale.

Consultant (Contract) at YLD

Jan 2017 – Apr 2017

Node.js, MySQL, SQL

- Worked alongside the client's CTO to improve the logging, tracing and error handling functions of an in-house Kubernetes-based microservices framework.
- Implemented 4 Node.js microservices while also supporting the team by developing shared helpers to mock data during test execution and writing documentation on how to migrate an SQLite database to MySQL.

External Advisor (Contract) at McKinsey & Company

Apr 2016 – Dec 2016

JavaScript, Python, React

- Counseled enterprise clients on software architecture, provided mentoring and conducted rigorous code reviews.
- Created a set of data analysis tools with visualisations built using D3.
- Engineered software to optimize energy and material usage within factories. This also involved implementing algorithms for calculating costs and analysing flows of energy, product, and waste through the system.

Tech Lead (Contract) at Home Office

Jan 2015 – Sep 2015

Node.js, JavaScript

- Re-platformed a legacy monolithic passport system into a Node.js microservices solution on a private cloud, incorporating secure data segregation, idempotent APIs, and custom LDAP authentication. Managed a small dedicated team throughout this process.
- Presented a new platform architecture to the entire department during a show-and-tell.

Previous Client Work

Apr 2013 – Apr 2016

- Development of a real-time animated map of driver locations for Hailo, an [open-source analytics middleware for the Sequoia-backed startup Keen.IO](#), and a React website and component library for the Economist.

We R Interactive

Lead Game Developer

Oct 2012 – Apr 2013

- Delivered MVP of an innovative second-screen social game utilising Node.js, WebSockets, Redis and Cassandra.

Saffron Digital

Lead Python Developer

Oct 2010 – Oct 2012

- Founding back-end engineer on the HTC Watch project, leading development until HTC's acquisition for \$48m.
- Developed expertise in DRM, parallel video encoding, and payment services using Python, MySQL and AWS.
- Developed an application for a Samsung Connected TV device with challenging hardware limitations.

Education

University of Kent

Bachelor's degree in Computer Science

2005 – 2009